Coding & Solution

Date	30 June 2026
Team ID	xxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/grandhisrisujan5-sys/EduGenie-Gemini-Learning-Assistant
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	FastAPI, Uvicorn, Jinja2, Google Gemini API, Hugging Face Transformers, PyMuPDF, Pillow

Key Features Implemented	AI Question Answering , Text Summarization , Quiz Generation Learning Roadmap Generator , PDF Chat & Summarization Image Analysis , File Upload Support
Pending / Incomplete Features	User Authentication , Database Integration , Cloud Deployment User Profile Management
Setup / Run Instructions	Install dependencies using requirements.txt. Configure the Gemini API key. Run the FastAPI server using Uvicorn. Open localhost:8000 in the browser. Upload PDFs or images and use the AI features.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The EduGenie – Gemini Learning Assistant is developed using FastAPI and Google Gemini API to provide AI-powered educational assistance. The project follows a modular architecture with separate frontend, backend, and service modules, making the code easy to understand and maintain. Features such as question answering, text summarization, quiz generation, learning roadmap creation, PDF analysis, and image understanding have been successfully implemented. The complete source code is maintained in a GitHub repository using version control, enabling efficient collaboration and future enhancements